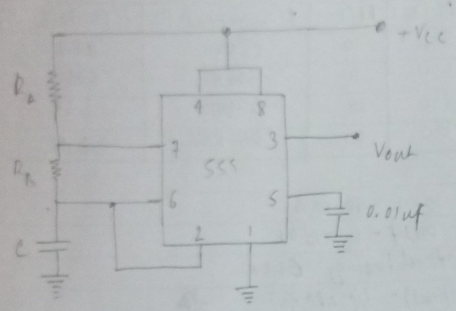
Circuit Diagram:



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Title:- Simulation of Astable Multivibrator Using IC 555 (using tinkercad / proteus).

Theory:-

The astable multivibrator has no Stable State it free runs, continuously switching between HIGH and LOW. This generates a rectangular wave at the output (pin 3) without requiring any external trigger.

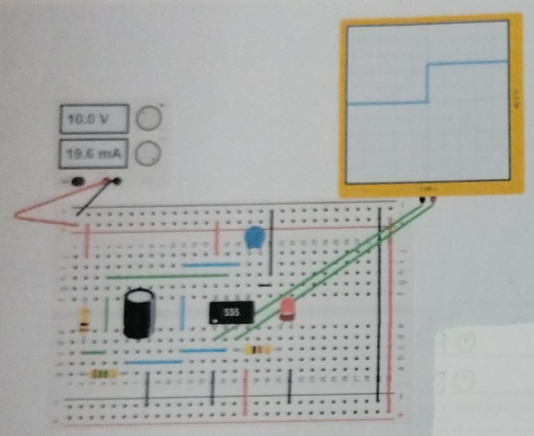
- Charging time (output High): $t_1 = 0.693 \times (R_A + R_B) \times C$
- Discharging time (output low): $t_2 = 0.693 \times R_B \times C$
- Time Period: $T_0 = t_1 + t_2 = 0.693 \times (R_A + 2R_B) \times C$
- Frequency: $f = 1/T = 1.44/(R_A + 2R_B) \times C$
- Duty cycle: % Duty Cycle = $[R_A + R_B / R_A + 2R_B] \times 100$

The oscillation is governed by the charging and discharging cycles of an external timing capacitor connected between pins 2/6 and ground.

Component Required

SrN	Name	Specification	Quantity
1	Timer IC	555	1
2	DC Power Supply	10.0v	2
3	Function Generator	1kHz, 2V, 0v	1
4)	Multimeter	digital	1
5	Capacitor	Polarized electrolytic	2
6)	LED	Red	1

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Observation Table :-

SN	R_A (kΩ)	R_B (kΩ)	C (nF)	f (Theoretical) Hz	f (Practical) Hz	Duty Cycle (%)
1	10	10	10	4600	5000	66.67%
2	16	4.7	0.01	7423	7500	75.77%
3	1	10	0.01	6857	7000	52.38%
4	10	10	0.1	480	495	66.67%

Calculation :-

$f(\text{theoretical}) = 1.44 / [(R_A + 2R_B) \times C]$

$= 1.44 / [(10 \times 10^3 + 2(10 \times 10^3)) \times (0.1 \times 10^{-6})]$

$= 480\ Hz$

$\text{Duty Cycle} = [(R_A + R_B) / (R_A + 2R_B)] \times 100$

$= [(10+10)/(10+20)] \times 100$

$= 66.67\%$

$\% \text{ Error} = \frac{[\text{Theoritical} - \text{Practical}]}{\text{Theoritical}} \times 100$

$= \frac{480 - 495}{480} \times 100$

$= 3.125\%$

Conclusion :-

The Simulation of the astable multivibrator using the IC 555 timer was succesfully conducted. By comparing the data in the observation table, it is evident that the observation table, it is evident that the theoretical frequency calculations closely match the practical/measured simulation frequencies. The experiment effectively verifies the formulas for determining the frequency and duty cycle.

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Data Sheet

EXPERIMENT NO. 11B

Title: - Simulation of Monostable Multivibrator using IC 555 (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	Breadboard	Small	1
2	Timer IC	555	1
3	Resistor	10KΩ, 100KΩ, 47KΩ	3
4	Capacitor	0.1uf, 0.01 uf	2
5	Power Supply	10V	1
6	Function Generator	1KHz, 2V, 0V	1
7	Multimeter	5V, 5A	1

Important Formulae: -

Operating Principle: -

The monostable (one-shot) multivibrator has one stable state. A negative trigger pulse at pin 2 (below V_{CC} /3) switches the output HIGH for a fixed time T, after which it automatically returns to the stable LOW state.

Output Pulse Width: -

$T = 1.1 \times R \times C$

- R = Timing resistor (connected between pin 7 and V_{CC})
- C = Timing capacitor (connected between pin 6/7 and ground)
- The capacitor charges through R until it reaches (2/3) V_{CC}, at which point the output resets to LOW

Observation Table: -

Sl. No.	R (kΩ)	C (μF)	Pulse Width, T (Theoretical) ms	Pulse Width, T (Practical / Measured) ms
1.	10	0.1	1.10	1.15
2.	10	0.01	0.11	0.12
3.	100	0.1	11.00	11.20

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Title :- Simulation of Monostable Multivibrator Using IC 555 (Using Tinkercad / Proteus).

Theory :-

The monostable or "One-Shot", multivibrator is a circuit that possesses exactly one stable state. The circuit remains in this stable LOW state until it is triggered. When a negative trigger pulse is applied at pin 2 it forces the output to switch to a High state for a fixed, predetermined time duration(T).

After this time T elapses the output automatically returns to its Stable Low State.

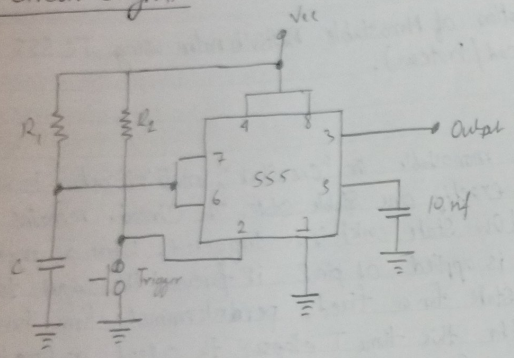
The width of this generated output pulse is determined exclusively by the external resistor and Capacitor using the formula: $T = 1.1 \times R \times C$.

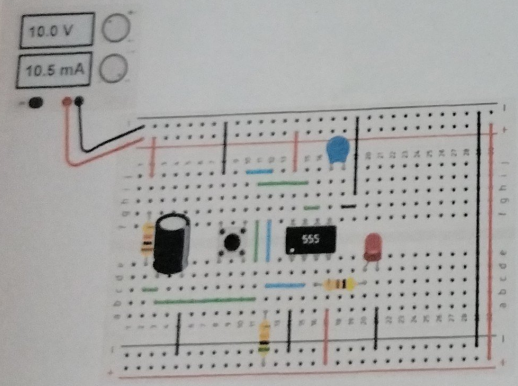
Component Required :-

SN	Name	Specification	Quantity
1	Breadboard	Small	1
2	Timer IC	555	1
3	Timing Resistor(R)	10KΩ, 100KΩ, 47KΩ	~~As required~~ 3
4	Timing Capacitor	0.1uf, 0.01uf	2
5	Power Supply	10V	1
6	Function Generator	1KHz, 2V, 0V	1
7	Multimeter	5V & 5A	1

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Circuit Diagram:-





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Observation Table:-

SN	R (KΩ)	C (μF)	Pulse Width, T (Theoretical) ms	Pulse Width, T (Practical) ms
1	10	0.1	1.10	1.15
2	10	0.01	0.11	0.12
3	100	0.1	11.00	11.20
4	47	0.1	5.17	5.30

Calculation:-

R = 10KΩ, C = 0.1 μF, Practical Pulse Width = 1.15ms

T(Theoretical) = $1.1 \times R \times C = 1.1 \times 10 \times 0.1 = 1.10$ ms.

% Error = $\left(\frac{\text{prac} - \text{theo}}{\text{prac}}\right) \times 100\ \% = \left(\frac{1.15 - 1.10}{1.10}\right) \times 100 = 4.55\ \%$

Calculation Conclusion:-

The Stimulation of the monostable multivibrator using the IC 555 timer was succesfully executed. By comparing the results in the observation table, it is clear that the theoretical pulse widths calculated using the formula $T = 1.1 \times R \times C$ align closely with the practical measured values.

Yielding a minimal error of 4.55%, this succesfully validates the operating principal of the one shot timer circuit.

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